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Institutional investor stability, executive equity incentives, and corporate innovation

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ABSTRACT

This study utilizes financial data from listed companies between 2013 and 2022 to examine the relationship between institutional investor stability and corporate innovation. The results indicate that institutional investor stability enhances corporate innovation levels. This positive effect exhibits heterogeneity across firms with varying levels of managerial control and different scales. Furthermore, institutional investor stability promotes corporate innovation by increasing executive equity incentives, while the increase in agency costs weakens the positive effect of institutional investor stability on corporate innovation.

1. Introduction

Amid intensifying global economic competition and rising uncertainties, corporate innovation has become a core driver for enhancing national and enterprise competitiveness. Internationally, developed economies such as the United States and Germany have been actively fostering innovation ecosystems by optimizing capital markets and corporate governance systems (Bolor-Erdene et al., 2023). Among these efforts, stable institutional investors provide crucial support for corporate innovation through sustained capital investment and alleviation of short-term performance pressures (Cella, 2020). Simultaneously, executive equity incentives, widely employed across firms in various countries, serve as an important tool for motivating management to pursue innovative endeavors. However, the instability in institutional investor behavior caused by increasing capital mobility, along with executives' inclination toward short-term decision-making, has emerged as a critical challenge in global corporate innovation governance, particularly in emerging economies (García-Sánchez et al., 2024). In China, the rapid expansion of capital markets has led to a significant increase in the scale of institutional investors, whose investment behaviors have drawn considerable attention regarding their impact on corporate innovation. Compared to developed economies, China's capital markets exhibit higher levels of uncertainty, with institutional investors demonstrating a mix of short-term and long-term investment preferences. At the same time, the executive equity incentive mechanism is still in the process of gradual improvement, and how to release the potential for corporate innovation through effective incentives has become an urgent issue to be resolved.

The academic community has produced a wealth of research on the stability of institutional investors, equity incentives for executives, and corporate innovation. From the perspective of institutional investor stability, Cella (2020) emphasizes the role of

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long-term institutional investors in mitigating agency problems and improving investment efficiency, highlighting the importance of institutional investor stability for corporate policies and performance. Kaldonski et al. (2020) argue that institutional investor stability plays a critical role in corporate governance and earnings management, as it helps to reduce real earnings management practices by managers aimed at meeting short-term performance targets. Li and Jiang (2022) offer new insights into institutional investor stability and its impact on corporate governance and market efficiency, particularly in emerging markets like China, where the influence of institutional investor networks is especially significant. Saci and Jasimuddin (2021) explore the impact of institutional investor research activities on the equity costs of Chinese listed companies, demonstrating that institutional investor stability enhances the fairness and transparency of information disclosure and reduces corporate equity costs. Regarding executive equity incentives, extensive studies have confirmed that equity incentives improve executives' willingness to take risks and their enthusiasm for innovation (Lahlou and Navatte, 2017; Zhang et al., 2024). For instance, the research of Zhang and Li (2024) demonstrates that equity incentives effectively stimulate executives to focus on long-term corporate value. However, some studies also indicate that executives' risk tolerance may be influenced by external factors such as capital market uncertainty and regulatory environments, which could affect the effectiveness of equity incentives (Chapman and Hewitt-Dundas, 2018; Cheng et al., 2023). From the perspective of corporate innovation, it is a critical factor in enhancing competitiveness and driving national economic growth. Bai et al. (2024) find that supply chain financing significantly promotes corporate innovation, and this effect is more pronounced in the context of inclusive digital financial development. Yang et al. (2024) examine the impact of administrative monopolies on corporate innovation, revealing that administrative monopolies significantly enhance innovation levels through resource access and market competition mechanisms. Yang et al. (2024) investigate the relationship between industry and regional peer effects and corporate innovation, concluding that firms' innovation behaviors are significantly influenced by industry and regional peer dynamics.

A review and synthesis of the aforementioned literature reveal that while substantial research has been conducted on institutional investor stability, executive equity incentives, and corporate innovation individually, studies integrating these three themes remain relatively scarce. This paper addresses this gap by examining the impact of institutional investor stability on corporate innovation, with executive equity incentives as a mediating variable. The study also examines the role of agency costs in the relationship between institutional investor stability and corporate innovation. The results show that institutional investor stability significantly fosters corporate innovation, particularly in firms with strong internal controls and larger organizations. Mediation analysis reveals that institutional investor stability promotes corporate innovation by increasing executive equity incentives. Additionally, moderation analysis indicates that higher agency costs diminish the positive effect of institutional investor stability on corporate innovation.

This study theoretically enriches the research on internal and external governance mechanisms of corporate innovation and provides practical guidance for designing more effective incentive mechanisms and optimizing shareholder structures. It also offers valuable insights for policymakers to improve the governance framework of capital markets and enhance corporate innovation capabilities. By uncovering the multifaceted roles of internal and external governance in corporate innovation, this study contributes to fostering long-term corporate competitiveness and promoting high-quality economic development.

2. Theoretical analysis and research hypotheses

2.1. Institutional investor stability and corporate innovation

Institutional investor stability, as a critical feature of external governance, is considered to play a significant role in promoting corporate innovation. According to agency theory, stable institutional investors, through long-term shareholding, provide clear incentives and constraints for management, encouraging them to focus on the firm's long-term development rather than short-term performance (Zhang et al., 2024). Innovation activities are often associated with high risks and uncertainties, and management may reduce R&D investments due to market pressures. Stable institutional investors, however, can alleviate this short-term orientation by offering sustained capital support and oversight of decision-making, thereby ensuring the continuous advancement of innovation activities. Moreover, long-term institutional investors develop a deep understanding of the firm's operations, enabling them to provide high-quality informational support to management and reduce information asymmetry. This role not only improves decision-making efficiency but also enhances the firm's credibility in capital markets, strengthening its financing capacity and laying a solid resource foundation for innovation activities (Kałdoński and Jewartowski, 2024). Additionally, behavioral finance theory highlights that stable institutional investors act as "stabilizers" during market fluctuations, helping firms maintain consistent R&D investments, which in turn enhances innovation efficiency and the conversion rate of innovation outcomes (Altaf and Jan 2023). Building on the previous findings, this study puts forward the following hypothesis:

Hypothesis 1. Institutional investor stability has a significant positive impact on corporate innovation.

2.2. Institutional investor stability, executive equity incentives, and corporate innovation

Corporate innovation is a crucial pathway for enhancing core competitiveness and achieving sustainable development. However, its characteristics of high investment, high risk, and high uncertainty make effective external governance and internal incentives particularly important (Belghitar and Clark, 2015). As a key factor in external governance, institutional investor stability provides firms with sustained capital support through long-term shareholding while establishing stable supervision and constraints on management. Internally, executive equity incentives serve as a vital mechanism to stimulate innovation motivation by aligning management interests with the firm's long-term goals, thereby enhancing executives' willingness to commit to innovation projects. Equity

incentives not only encourage executives to take on the high risks associated with innovation but also drive their focus on R&D and technological upgrades, facilitating the realization of long-term corporate value (Huang et al., 2025; Lai et al., 2017). Additionally, equity incentives help firms attract and retain core managerial talent, which is particularly crucial for knowledge-intensive and technology-oriented enterprises. Specifically, institutional investor stability offers external assurance for corporate innovation through sustained support and governance optimization. Meanwhile, executive equity incentives provide internal momentum for innovation through multi-level incentive mechanisms. Together, they complement each other, jointly enhancing corporate innovation capabilities and laying a solid foundation for long-term competitiveness. Based on the above analysis, this study proposes the following hypothesis:

Hypothesis 2. Institutional investor stability enhances corporate innovation by increasing executive equity incentives.

2.3. Institutional investor stability, agency costs, and corporate innovation

Agency costs refer to the efficiency losses arising from information asymmetry, conflicting objectives, and other factors between corporate management and shareholders, which are particularly pronounced in the context of corporate innovation activities (Belghitar and Clark, 2015). High-stability institutional investors provide long-term capital support and foster greater focus on long-term development and innovation through effective supervision mechanisms. However, as agency costs increase, the positive effect of institutional investor stability on innovation tends to weaken. High agency costs signify more pronounced information asymmetry and conflicts of interest between management and shareholders. Under such conditions, management may be inclined to avoid high-risk innovation projects and favor conservative strategies with short-term returns (Zhang and Li, 2024). This weakens the supervisory and incentivizing effects of institutional investors, thereby diminishing their positive impact on corporate innovation. In light of the previous analysis, the following hypothesis is proposed in this study:

Hypothesis 3. The increase in agency costs weakens the positive effect of institutional investor stability on corporate innovation.

3. Research design

3.1. Data sources

This study analyzes A-share listed companies from 2013 to 2022, using data sourced from the CSMAR database. Companies categorized as ST and *ST, as well as those in the financial sector, were excluded from the sample. Furthermore, observations with incomplete data were omitted, and all continuous variables underwent winsorization. The final dataset includes 11,759 observations.

3.2. Variable definitions

Dependent variable: Corporate innovation (*RD*). Innovation is the primary driver of development, directly influencing a firm's sustainable growth and its ability to gain a competitive edge in intense market competition, thereby enhancing overall competitiveness (Bai et al., 2024). This study uses firms' R&D expenditure to measure their level of innovation. R&D expenditure forms the foundation of corporate innovation activities, encompassing direct financial investments in new products, new services, and technological improvements. In the robustness tests, the number of annual patent applications is used as an alternative measure of corporate innovation..

Independent variable: Institutional investor stability (*IO*). Following the methodologies of Callen and Fang (2013) and Wang and Sun (2022), this study measures institutional investor stability based on the proportion of institutional shareholding, calculated from a temporal perspective. As shown in Eq. (1), institutional investor stability is defined as the ratio of the institutional shareholding proportion of firm *i* in year *t* to the standard deviation of institutional shareholding proportions for firm ii over the previous three years (from $t-1$ to $t-3$).

$$SD_{it} = \frac{INVH_{it}}{STD(INVH_{it-3}, INVH_{it-2}, INVH_{it-1})} \tag{1}$$

Where $INVH_{it}$ represents the proportion of institutional shareholding for firm *i* in year *t*, $STD(INVH_{it-3}, INVH_{it-2}, INVH_{it-1})$ denotes the standard deviation of institutional shareholding proportions for firm *i* over the preceding three years, and SD_{it} is the calculated value of institutional investor stability. A higher SD_{it} indicates greater stability of institutional investors.

Mediating variable: Executive equity incentives (*MSR*). Building on existing literature and considering data availability, this study measures executive equity incentives using the natural logarithm of the number of shares held by executives plus one. A larger value indicates a higher level of executive equity incentives (Li and Kuo, 2017; Zeng et al., 2023).

Moderating variable: Agency costs (*AC*). Following relevant theories (Varela, 2017) and drawing on the studies by Schäuble (2019) and Lin et al. (2020), this study uses the administrative expense ratio to measure agency costs. The administrative expense ratio is calculated as the proportion of administrative expenses to operating revenue. A higher administrative expense ratio suggests greater agency costs between shareholders and management.

Control variables. To capture other factors that may impact corporate innovation and improve the model's explanatory power, this study incorporates the following control variables based on relevant literature (Wang et al., 2023): leverage ratio (*Lev*), return on equity (*Roe*), cash flow ratio (*Cashflow*), fixed asset ratio (*FIXED*), revenue growth rate (*Growth*), audit quality (*Big4*), total asset

turnover (*ATO*), and Tobin's Q value (*TobinQ*), as shown in Table 1.

3.3. Model specification

In order to examine the link between institutional investor stability and corporate innovation, this study designates corporate innovation as the dependent variable and institutional investor stability as the independent variable. The following model is proposed:

$$RD_{it} = \alpha_0 + \alpha_1 IO_{it} + Control_{it} + \mu_i + \gamma_t + \varepsilon_{it} \tag{2}$$

Where the subscript i denotes the firm, and t represents the time period. RD_{it} measures the innovation level of firm i at time t. IO_{it} represents institutional investor stability, and $Control_{it}$ is the set of control variables. μ_i and γ_t denote industry and time fixed effects, respectively, while ε_{it} is the exogenous disturbance term.

For Hypothesis 2, this study introduces *MSR* to construct a mediation effect model, as shown in Eqs. (3) and (4).

$$RD_{it} = \beta_0 + \beta_1 IO_{it} + Control_{it} + \mu_i + \gamma_t + \varepsilon_{it} \tag{3}$$

$$RD_{it} = \theta_0 + \theta_1 IO_{it} + \theta_2 MSR_{it} + Control_{it} + \mu_i + \gamma_t + \varepsilon_{it} \tag{4}$$

Where MSR_{it} represents executive equity incentives.

For Hypothesis 3, this study incorporates *AC* and its interaction term with *IO* to establish a moderation effect model, as presented in Eq. (5).

$$RD_{it} = \pi_0 + \pi_1 IO_{it} + \pi_2 AC_{it} + \pi_3 IO_{it} \times AC_{it} + Control_{it} + \mu_i + \gamma_t + \varepsilon_{it} \tag{5}$$

Where AC_{it} represents agency costs, and $IO_{it} \times AC_{it}$ denotes the interaction term between institutional investor stability and agency costs.

4. Empirical analysis

4.1. Descriptive analysis

Table 2 presents the descriptive statistics for the data used in this study. The standard deviation of *RD* indicates some differences across firms, though the overall fluctuation is relatively moderate. The median suggests a relatively symmetric distribution of the data. The average value of *IO* reflects a moderately high level among sample firms, while the standard deviation suggests significant variation across firms. The maximum and minimum values of *MSR* reveal a wide range, which may be related to differences in firms' ownership structures and incentive policies. The median value is slightly higher than the mean, indicating a possible right-skewed distribution, with a subset of firms exhibiting particularly high *MSR*. The standard deviation of *AC* reflects a certain degree of dispersion. The values of *AC* highlight substantial differences across firms. Overall, these indicators demonstrate the diversity and moderate central tendency in *RD, IO, MSR*, and *AC* among firms, providing a solid foundation for subsequent empirical analysis.

4.2. Baseline regression

Table 3 presents the baseline regression results. Column (1) shows the results without control variables, where the impact of *IO* on *RD* is statistically significant at the 1 % level, indicating that *RD* has a significantly positive effect on *IO*, thus validating Hypothesis 1. In Column (2), after incorporating a series of control variables, the results remain positively significant at the 1 % level. This demonstrates that even when accounting for other potential factors influencing *RD, IO* continues to exert a significantly positive effect on *RD*. Specifically, for every one standard deviation increase in institutional investor stability, corporate innovation increases by 10.02

Table 1
Variable definitions.

Variable type	Variable name	Symbol	Definition
Dependent variable	Corporate innovation	*RD*	Log-transformed R&D investment plus 1.
		PT	Log-transformed total annual patent applications plus 1.
Independent variable	Institutional investor stability	*IO*	Calculated according to Eq. (1).
Mediating variable	Executive equity incentives	*MSR*	Log-transformed number of shares held by management plus 1.
Moderating variable	Agency costs	*AC*	Administrative expenses / operating revenue.
Control variables	Debt-to-Asset ratio	*Lev*	Total liabilities / total assets.
	Return on equity	*Roe*	Net profit / average net assets.
	Cash flow ratio	*Cashflow*	Net cash flow from operating activities / total assets.
	Fixed asset ratio	*FIXED*	Net fixed assets / total assets.
	Revenue growth rate	*Growth*	(Current year's revenue / Previous year's revenue) - 1.
	Audit quality	*Big4*	Equal to 1 if audited by big four (PwC, Deloitte, KPMG, EY), otherwise 0.
	Asset turnover ratio	*ATO*	Operating revenue divided by average total assets of the current and previous year.
	Tobin's Q	*TobinQ*	Firm's market value divided by total assets.

Table 2
Summary statistics.

VarName	Obs	Mean	SD	Min	Max
RD	11,759	18.3176	1.3032	14.9731	22.0448
PT	11,759	2.0646	2.1764	0.0000	6.9976
IO	11,759	2.4292	1.2223	0.1022	5.8576
MSR	11,759	12.5426	6.6763	0.0000	19.9874
AC	11,759	0.0863	0.0642	0.0068	0.6409
Lev	11,759	0.4104	0.1867	0.0645	0.8542
ROE	11,759	0.0617	0.1166	−0.4794	0.3319
Cashflow	11,759	0.0498	0.0615	−0.1129	0.2265
FIXED	11,759	0.2075	0.1339	0.0083	0.6030
Growth	11,759	0.1613	0.3318	−0.4644	1.7399
Big4	11,759	0.0451	0.2075	0.0000	1.0000
ATO	11,759	0.6255	0.3589	0.1259	2.2256
TobinQ	11,759	2.1672	1.2969	0.8619	7.6868

Table 3
Baseline regression.

	(1) *RD*	(2) *RD*
IO	0.1603***	0.0820***
	(6.5444)	(6.0822)
Lev		1.8114***
		(8.9780)
ROE		2.1829***
		(12.9620)
Cashflow		1.9984***
		(7.9113)
FIXED		−0.4413**
		(−2.0255)
Growth		−0.0882*
		(−1.9319)
Big4		1.1379***
		(7.3863)
ATO		0.5254***
		(6.7075)
TobinQ		−0.1730***
		(−8.0677)
_cons	17.9282***	17.2414***
	(301.3521)	(169.4495)
Ind FE	Yes	Yes
Year FE	Yes	Yes
N	11,759	11,759
Adj. R^2	0.1871	0.3953

Note: *** indicates a significance level of $P< 0.01$, ** indicates $P< 0.05$, and * represents $P< 0.1$. The t-values are presented in parentheses. FE stands for fixed effects The same applies to the following table.

%. This further corroborates Hypothesis 1.

4.3. Robustness tests

To further verify the reliability and validity of the research results, this study conducts robustness tests by replacing the dependent variable and excluding specific samples for regression analysis.

4.3.1. Replacing the dependent variable

To ensure that the research results are not dependent on a specific measurement indicator, this study uses alternative measures of corporate innovation. Specifically, the original dependent variable *RD* is replaced with *PT* (patent applications). R&D investment may be influenced by various factors, such as firm size, industry characteristics, and R&D efficiency, which could affect the relationship between R&D investment and corporate innovation. By using the number of patent applications as a proxy for innovation, this study mitigates potential biases caused by these factors and provides a complementary perspective for measuring innovation. Additionally, patent applications enable a better comparison of innovation levels across firms and industries, enhancing the generalizability and comparability of the results. The results, as shown in Table 4, indicate that after replacing the dependent variable, *IO* still has a significantly positive impact on *PT* at the 1 % level, further supporting Hypothesis 1.

4.3.2. Reducing the sample size

The 2015 stock market crash significantly disrupted the Chinese stock market, leading to a sharp decline in market value and a loss of investor confidence. Similarly, the 2020 COVID-19 pandemic triggered a global economic recession. To reduce the potential influence of outliers and improve the reliability and validity of the regression results, this study excludes data from these two exceptional years. The regression results, as presented in Table 4, demonstrate that the positive impact of *IO* on *RD* remains significant at the 1 % level, thereby validating the conclusions drawn for Hypothesis 1.

4.4. Endogeneity test

Given the long cycle of corporate innovation activities and the delayed effect of institutional investors on innovation, this study incorporates lagged regression models for robustness testing. The results are presented in Table 5. In Column (1), when *RD* is lagged by one period, the coefficient of *IO* on *RD* is 0.0780, which remains significant at the 1 % level. In Column (2), when *RD* is lagged by two periods, *IO* continues to exhibit a significant positive effect at the 1 % level. Similarly, Columns (3) and (4) present the regression results when *IO* is lagged by one and two periods, respectively, where the impact of *IO* on *RD* remains significant at the 1 % level. The results indicate that as the lag period increases, the coefficient of *IO* on *RD* gradually decreases. This suggests that the impact of institutional investor stability on corporate innovation weakens over time. However, the long-term effect remains significant, demonstrating the robustness of the conclusions.

4.5. Mechanism analysis

The results of the main effect analysis indicate that institutional investor stability contributes to enhancing corporate innovation capability. Next, this study further explores how institutional investor stability affects corporate innovation.

4.5.1. Mediation effect

This study examines the mediating effect of MSR in the relationship between *IO* and *RD*. The results are presented in Table 6. Column (1) shows the baseline regression results, where *IO* has a significant positive impact on *RD* at the 1 % level. In Column (2), after introducing *MSR*, the results still indicate that *IO* has a significant positive impact on *RD* at the 1 % level. This demonstrates the presence of the mediating effect of *MSR*. In other words, *IO* promotes *RD* by enhancing *MSR*, thereby confirming Hypothesis 2.

4.5.2. Moderation effect

Table 7 introduces *AC* as a moderating variable and incorporates the interaction term $IO \times AC$ into Eq. (4). Column (1) presents the baseline regression results. In Column (2), the coefficient of *AC* is negative, indicating that *AC* is negatively correlated with *RD*. This suggests that an increase in *AC* suppresses *RD*. Moreover, the coefficient of the interaction term $IO \times AC$ is also negative and significant at the 1 % level. This finding implies that the increase in AC weakens the positive effect of *IO* on *RD*, indicating that *AC* negatively moderates the relationship between *IO* and *RD*. Thus, Hypothesis 3 is confirmed.

5. Further analysis

5.1. Internal control quality

As a critical tool for enterprise risk management and operational efficiency, the value of internal control (*IC*) has become increasingly prominent in today's complex and dynamic business environment. *IC* not only affects the effective execution of strategic goals but also directly influences financial stability and market competitiveness (Li et al., 2024). Based on the sample data, this study conducts regressions by grouping firms according to the median value of *IC* and tests for differences in coefficients between groups. The results are shown in Table 8. In the Low *IC* group, the coefficient of *IO* on *RD* is 0.0602, while in the High *IC* group, the coefficient is 0.0972. Both coefficients are significant at the 1 % level. According to the coefficient difference test, the positive effect of *IO* on *RD* is more pronounced in firms with high *IC*. High levels of *IC* are often accompanied by stricter management and supervision systems,

Table 4
Robustness checks.

	Replacing the dependent variable *PT*	Reducing the sample size *RD*
IO	0.1445***	0.0884***
	(5.5567)	(5.6236)
_cons	1.3572***	17.2932***
	(10.2655)	(163.7486)
Controls	Yes	Yes
Ind FE	Yes	Yes
Year FE	Yes	Yes
N	11,759	9256
Adj. R^2	0.0886	0.4001

Table 5
Endogeneity test.

	(1) L.*RD*	(2) *L2.RD*	(3) *RD*	(4) *RD*
IO	0.0780*** (5.9327)	0.0766*** (5.2608)		
L.*IO*			0.0804*** (6.2546)	
L2.IO				0.0785*** (5.4629)
_cons	17.3111*** (166.8849)	17.3101*** (150.8633)	17.3257*** (178.9966)	17.3918*** (171.7490)
Controls	Yes	Yes	Yes	Yes
Ind FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	8152	6420	8152	6420
Adj. R^2	0.3822	0.3659	0.3931	0.4014

Table 6
MSR mediation effects.

	(1) *RD*	(2) *RD*
IO	0.0820*** (6.0822)	0.1074*** (8.5100)
MSR		0.0167*** (5.1753)
_cons	17.2414*** (169.4495)	16.9275*** (149.1161)
Controls	Yes	Yes
Ind FE	Yes	Yes
Year FE	Yes	Yes
N	11,759	11,759
Adj. R^2	0.3953	0.4011

Table 7
Moderating effect of AC.

	(1) RD	(2) RD
IO	0.0820*** (6.0822)	0.0809*** (5.4622)
AC		−1.7194*** (−2.9787)
IO × *AC*		−0.8573*** (−6.2201)
_cons	17.2414*** (169.4495)	17.5105*** (127.1491)
Controls	Yes	Yes
Ind FE	Yes	Yes
Year FE	Yes	Yes
N	11,759	11,759
Adj. R^2	0.3953	0.3736

which not only effectively reduce operational risks but also enhance decision-making efficiency and transparency. Furthermore, high-quality *IC* ensures the reliability of financial reporting, lowers the cost of capital, and attracts more external investment. In such an environment, institutional investor stability provides more predictable and reliable external support, further boosting firms' confidence in undertaking long-term and high-risk innovation activities.

5.2. Firm size

Significant differences exist between firms of varying sizes in terms of resource allocation, managerial complexity, and market influence. Based on the median value of firm size, the sample is divided into two groups: small-to-medium-sized enterprises (SMEs) and large firms, and regressions are performed for each group. The results are shown in Table 8. The effect of *IO* on *RD* in large firms is positively significant at the 1 % level. Compared to SMEs, large firms, with their stronger capital base and more extensive resource

Table 8
Heterogeneity analysis.

	Low *IC* *RD*	High *IC* *RD*	SMEs *RD*	Large Enterprises *RD*
IO	0.0602***	0.0972***	−0.0092	0.0791***
	(4.3446)	(5.2529)	(−0.6909)	(3.4013)
_cons	17.5609***	17.0257***	17.5279***	17.5744***
	(186.1812)	(144.1605)	(242.7106)	(126.4787)
Controls	Yes	Yes	Yes	Yes
Ind FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	5879	5880	5879	5880
Adj. R^2	0.3670	0.4255	0.3003	0.3867
Fisher combined test p-values	0.0080		0.0000	

reserves, are able to allocate more funds to R&D activities and better absorb the potential risks associated with innovation failure. Additionally, differences in market influence between the two groups are noteworthy. Large firms, due to their size and resource advantages, often dominate the market, making it easier for their R&D outcomes to gain acceptance and be commercialized. In contrast, SMEs need to rely on differentiation strategies and niche markets to enhance their market influence and the success rate of their innovation commercialization. In this context, institutional investor stability plays a critical role by providing sustained capital support and governance optimization. This helps firms maintain an optimal scale for maximizing innovation returns, further improving innovation efficiency and the conversion of innovation outcomes into tangible benefits.

6. Conclusions and recommendations

Based on firm internal financial data, this study explores the relationship between institutional investor stability, executive incentives, and corporate innovation. The findings are as follows: (1) Institutional investor stability enhances corporate innovation levels. (2) Institutional investor stability promotes corporate innovation by increasing executive equity ownership. (3) The positive effect of institutional investor stability on corporate innovation diminishes as agency costs increase. (4) The incentivizing effect of institutional investor stability varies across firms of different sizes and management characteristics, being more significant in large-scale firms and those with higher levels of internal management. Based on these findings, the following recommendations are proposed:

First, improve internal control mechanisms. Strengthen internal governance and enhance the quality of internal controls to ensure transparency in decision-making and reliability in financial information. High-quality internal controls not only reduce operational risks but also create a more stable investment environment for institutional investors, thereby increasing their willingness for long-term shareholding and providing stronger support for corporate innovation. Second, optimize executive equity incentive mechanism. Align executives' personal interests with the company's long-term innovation goals through equity incentives. Such mechanisms can enhance executives' sense of responsibility and belonging, motivating them to focus more on the firm's long-term development and innovation activities, ultimately boosting the company's overall innovation capacity. Third, reduce agency costs. Implement effective measures to lower agency costs by optimizing corporate governance structures, increasing information transparency, and strengthening internal audits. By reducing information asymmetry and conflicts of interest between management and shareholders, the stability of institutional investors can be better leveraged to drive innovation. Fourth, adopt differentiated innovation strategies. The scale of a firm determines its resource endowment and strategic choices. Large firms should fully leverage their capital and resource advantages to increase R&D investment and attract stable institutional investors to strengthen long-term innovation capacity. Meanwhile, SMEs should enhance internal controls and actively integrate external resources to alleviate financial constraints through institutional investor support, thereby improving innovation efficiency. Fifth, optimize the capital market policy environment. Policymakers should improve regulatory mechanisms in the capital market and encourage institutional investors to adopt long-term shareholding strategies. Measures such as optimizing tax policies and enhancing information disclosure transparency can increase institutional investor stability. Additionally, policymakers should provide support for corporate innovation, such as establishing innovation incentive funds or targeted support programs, to guide firms toward efficient innovation within a stable investment environment.

Although this study reveals the positive effect of institutional investor stability on corporate innovation, there are still some limitations: First, sample data limitation. The sample data is confined to listed companies, without adequately considering differences in industries, regions, or ownership structures, which restricts the generalizability of the conclusions. Second, lack of dynamic analysis. The study lacks an analysis of the dynamic evolution of the relationship between institutional investor stability and corporate innovation, failing to capture its long-term effects. Future research could expand the sample data range and employ methods such as dynamic panel data models to enhance the generalizability and representativeness of the research conclusions.

CRediT authorship contribution statement

Haoyi Jia: Writing – review & editing, Data curation, Conceptualization. **Weina Che:** Writing – original draft, Methodology,

Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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